Phase 1: Research Paper Proposal (XAI Project)

Paper Title: InstaSHAP: Interpretable Additive Models Explain Shapley Values Instantly (ICLR 2025)

Paper Link: https://openreview.net/forum?id=ky7vVlBQBY

Problem Statement:

Computing exact Shapley Additive Explanations (SHAP) is theoretically optimal for feature attribution, but it is computationally expensive and too slow for real-time deployment. The process requires evaluating exponential feature permutations, making it impractical for large-scale applications.

Dataset Used:

The authors evaluated their approach on standard tabular benchmark datasets and the CUB (Caltech-UCSD Birds) image dataset.

XAI Method Used:

Surrogate Intrinsic Modeling. The authors trained Generalized Additive Models (GAMs) to instantly predict SHAP value distributions of a black-box model, enabling fast and interpretable explanations.

Reason for Selecting This Paper & Proposed Extension:

Standard GAMs assume feature independence and cannot effectively capture feature interactions. For our project, we propose replacing the standard GAM surrogate with an Explainable Boosting Machine (EBM), which models pairwise feature interactions while maintaining interpretability. By training EBM on the same tabular datasets used in the paper, we can compare accuracy and speed against the published GAM baseline and demonstrate improved SHAP approximation performance without sacrificing real-time efficiency.

Team Members:

- 1. J Ganesh Reddy : 23BDS024
- 2. M Jagadeeswar Reddy : 23BDS033
- 3. Aggimalla Abhishek : 23BDS004
- 4. LINGALA SAHITH AKASH MANIKANTA : 23BDS031
- 5. Shriyans Kandimalla : 23BCS064